A field-guide to Pseudopotentials and Relativistic Effects

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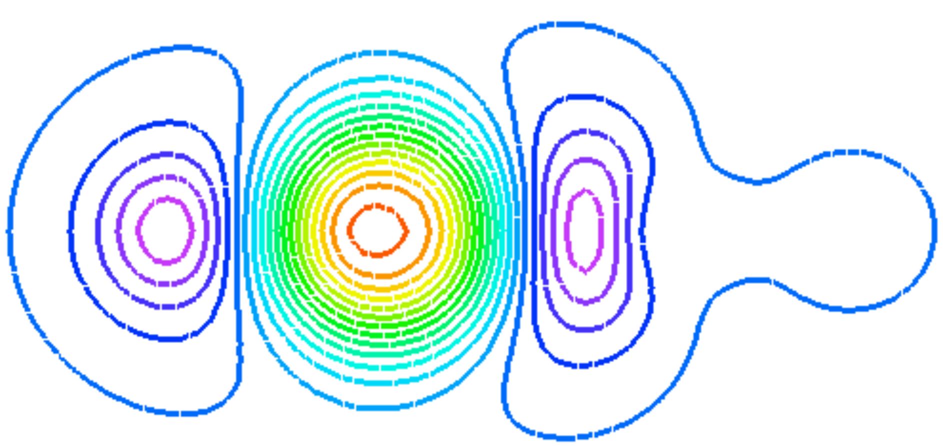
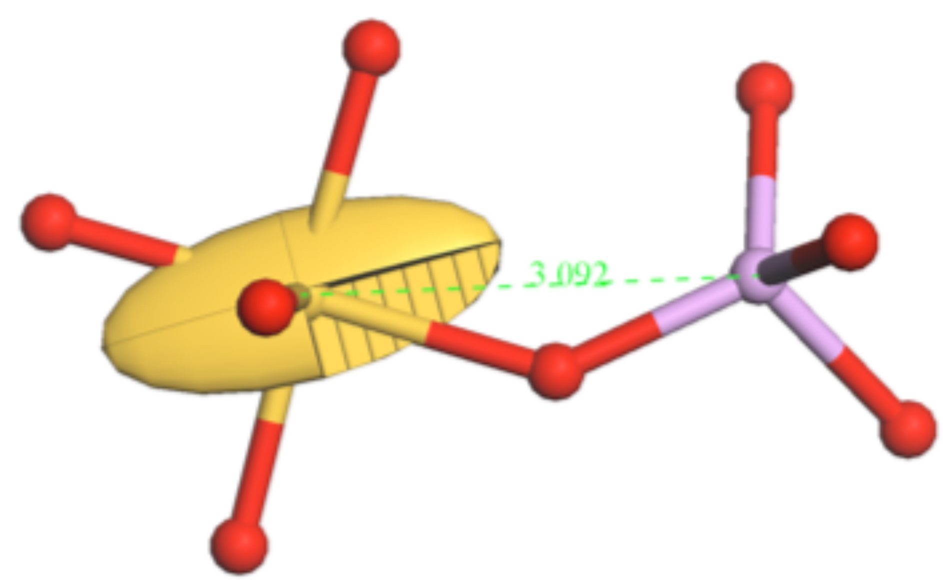




Planewave basis + pseudopotentials

Planewave Basis

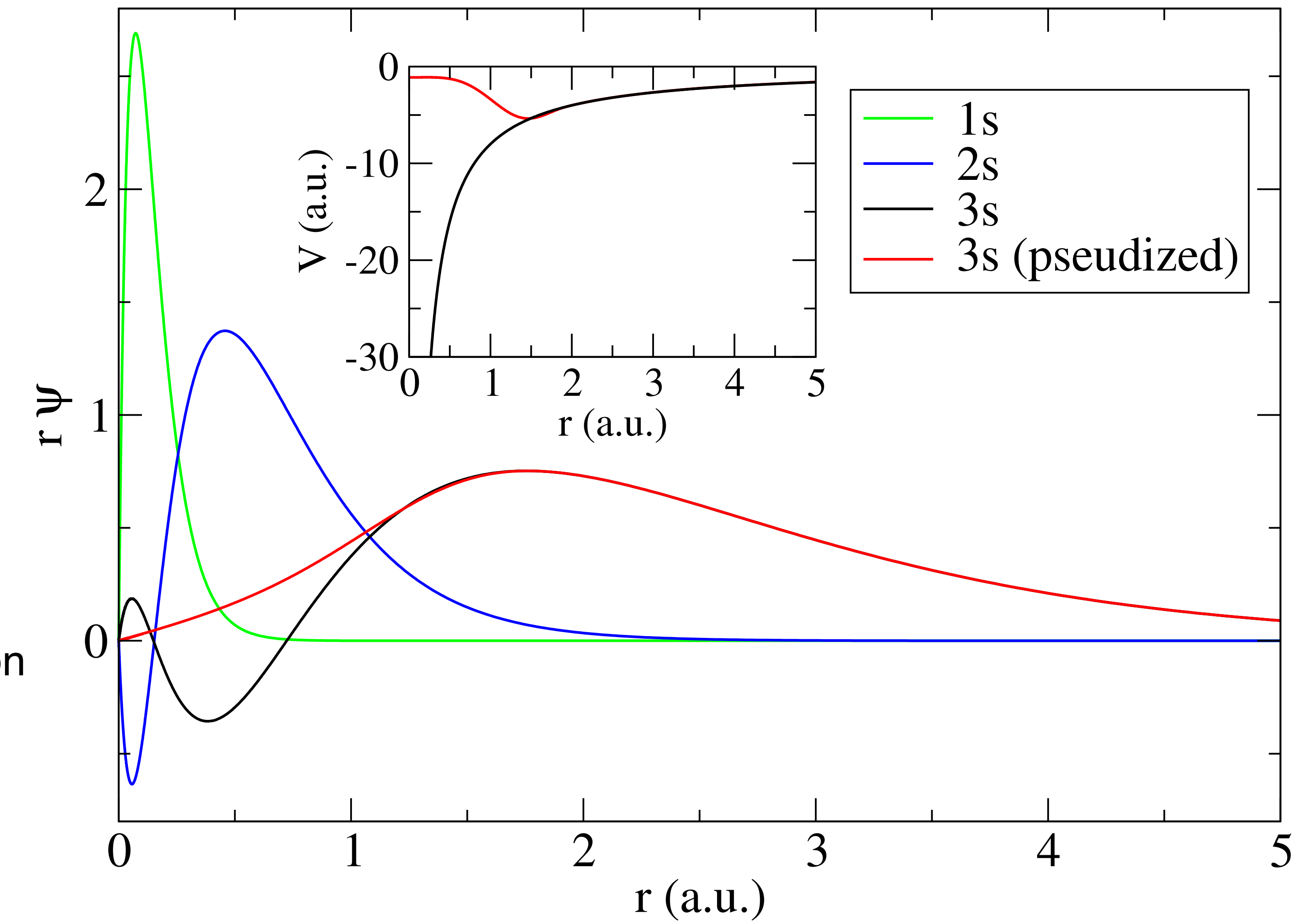
Pro
Satisfy Period Boundary conditions
Easy to check convergence

Con
Very expensive to treat core-electrons, and valence wavefunctions close to nucleus

Approximations

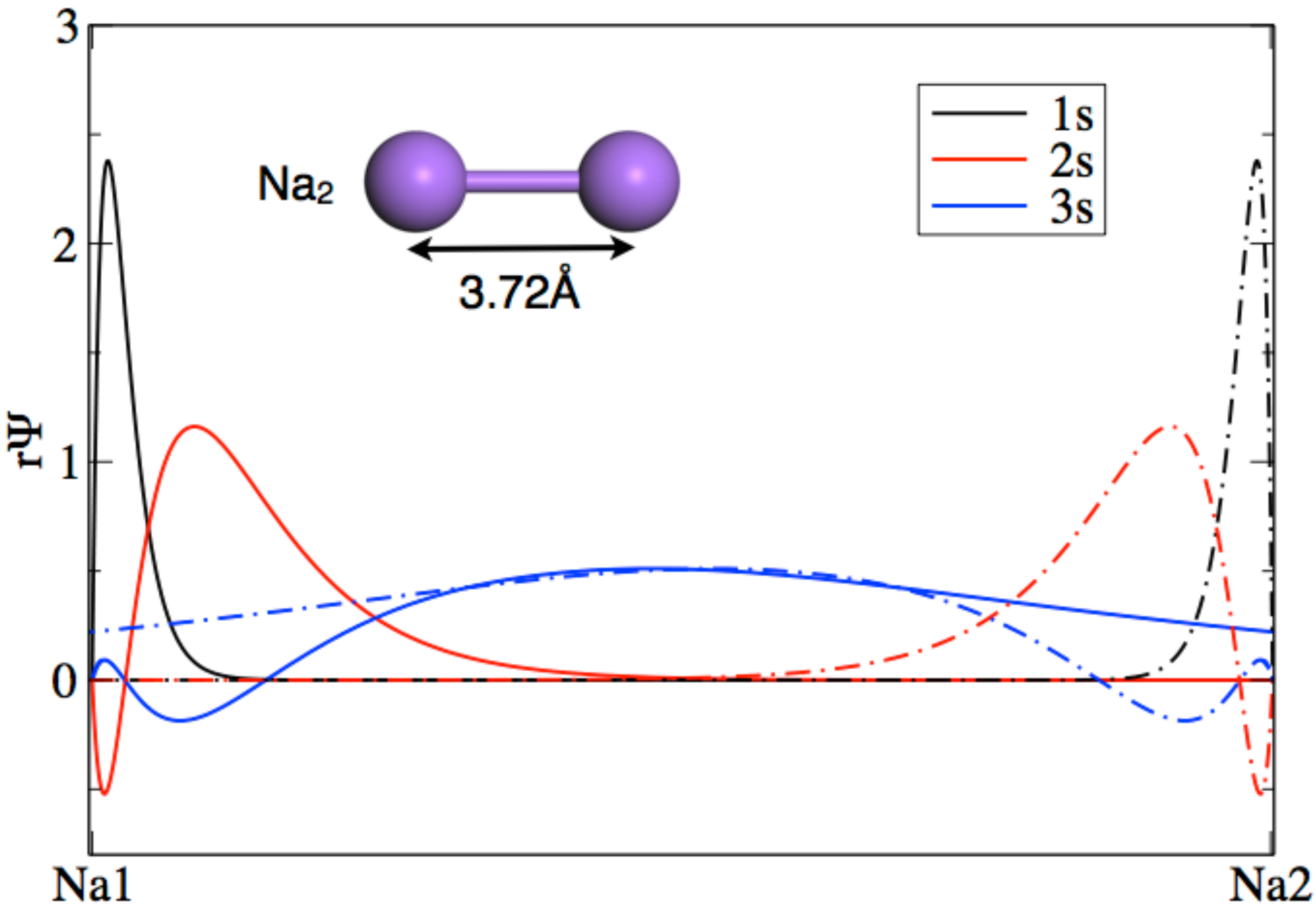
Frozen core
Take core states from a free atom calculation

Pseudopotential
Valence electron feel a shallow effective potential in core region



All good for bonding - problematic for NMR....

Sodium Dimer



Choosing Pseudopotentials

Things to consider

- Core radius (high pressure)
- Core vs valence - semi-core states

Norm-conserving / Ultrasoft

Ultrasoft considered “state-of-the-art” but

- Not always cheaper
- Not all properties implemented

Specifying Pseudopotentials

```
%BLOCK SPECIES_POT
C C_00PBE.usp
%ENDBLOCK SPECIES_POT
```

Potential from file - will look in $PSPOT_DIR (default to working directory)

```
%BLOCK SPECIES_POT
C 2|1.4|6|11|11|20:21(qc=7)
%ENDBLOCK SPECIES_POT
```

On-the-fly potential string

```
%BLOCK SPECIES_POT
NCP
%ENDBLOCK SPECIES_POT
```

On the fly norm-conserving library

```
%BLOCK SPECIES_POT
%ENDBLOCK SPECIES_POT
```

Use CASTEP's recommended USP

```
%BLOCK SPECIES_POT
C18
%ENDBLOCK SPECIES_POT
```

Use USP from CASTEPv18

```
%BLOCK SPECIES_POT
QC5
%ENDBLOCK SPECIES_POT
```

High-through put set

On the fly potentials

```
%BLOCK SPECIES_POT
C 2|1.4|6|11|11|20:21(qc=7)
%ENDBLOCK SPECIES_POT
```

```
============================================================
| Pseudopotential Report - Date of generation  9-09-2014   |
------------------------------------------------------------
| Element: C Ionic charge:  4.00 Level of theory: PBE      |
| Atomic Solver: Koelling-Harmon                           |
|                                                          |
|              Reference Electronic Structure              |
|        Orbital         Occupation         Energy         |
|           2s              2.000            -0.505         |
|           2p              2.000            -0.194         |
|                                                          |
|                Pseudopotential Definition                |
|       Beta      l      e       Rc      scheme   norm     |
|        1        0    -0.505   1.395     qc      0        |
|        2        0     0.250   1.395     qc      0        |
|        3        1    -0.194   1.395     qc      0        |
|        4        1     0.250   1.395     qc      0        |
|       loc       2     0.000   1.395     pn      0        |
|                                                          |
| Augmentation charge Rinner = 0.983                       |
| Partial core correction Rc = 0.983                       |
------------------------------------------------------------
| "2|1.4|6|11|11|20:21(qc=7)"                              |
------------------------------------------------------------
|      Author: Chris J. Pickard, Cambridge University      |
============================================================
```

On the fly potentials

```
%BLOCK SPECIES_POT
3|2.0|2.0|1.0|8|11|11|30U:40:31:32(qc=6)
%ENDBLOCK SPECIES_POT
```

```
============================================================
| Pseudopotential Report - Date of generation  9-09-2014   |
------------------------------------------------------------
| Element: V Ionic charge: 13.00 Level of theory: PBE      |
| Atomic Solver: Koelling-Harmon                           |
|                                                          |
|              Reference Electronic Structure              |
|        Orbital         Occupation         Energy         |
|           3s              2.000           -2.581         |
|           3p              6.000           -1.618         |
|           3d              3.000           -0.190         |
|           4s              2.000           -0.173         |
|                                                          |
|                Pseudopotential Definition                |
|       Beta      l      e        Rc     scheme   norm     |
|         1       0   -2.581    1.993     qc      0        |
|         2       0   -0.173    1.993     qc      0        |
|         3       0    0.250    1.993     qc      0        |
|         4       1   -1.618    1.993     qc      0        |
|         5       1    0.250    1.993     qc      0        |
|         6       2   -0.190    1.993     qc      0        |
|         7       2    0.250    1.993     qc      0        |
|       loc       3    0.000    1.993     pn      0        |
|                                                          |
| Augmentation charge Rinner = 0.998                       |
| Partial core correction Rc = 0.998                       |
------------------------------------------------------------
| "3|2.0|2.0|1.0|8|11|11|30U:40:31:32(qc=6)"               |
------------------------------------------------------------
|     Author: Chris J. Pickard, Cambridge University       |
============================================================
```

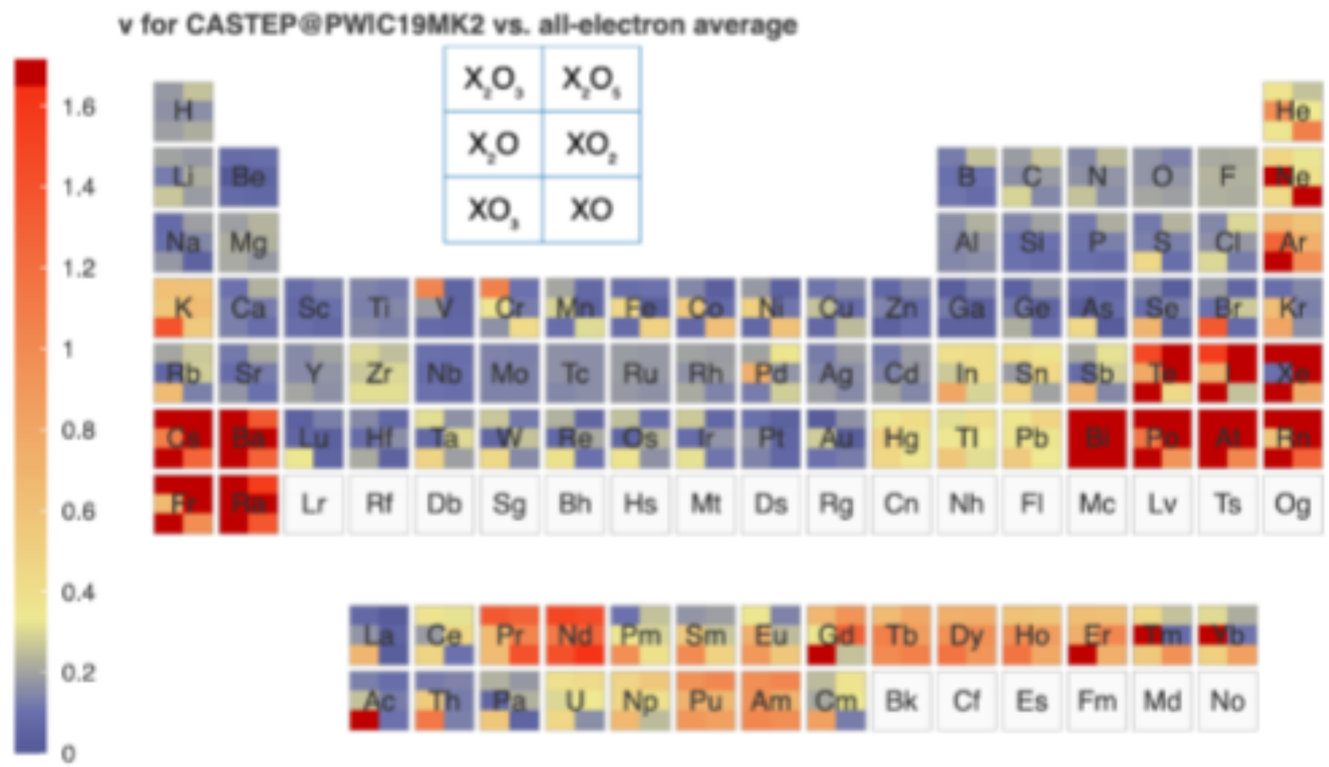
Pseudopotentials

http://molmod.ugent.be/deltacodesdft

Pseudopotentials for f-block elements

Follow-on work from the “delta” project extended tests to include oxides in test set and Lanthanides and Actinides.

New CASTEP OTF library “C19MK2” (identical to C19 for s,p,d-block elements).



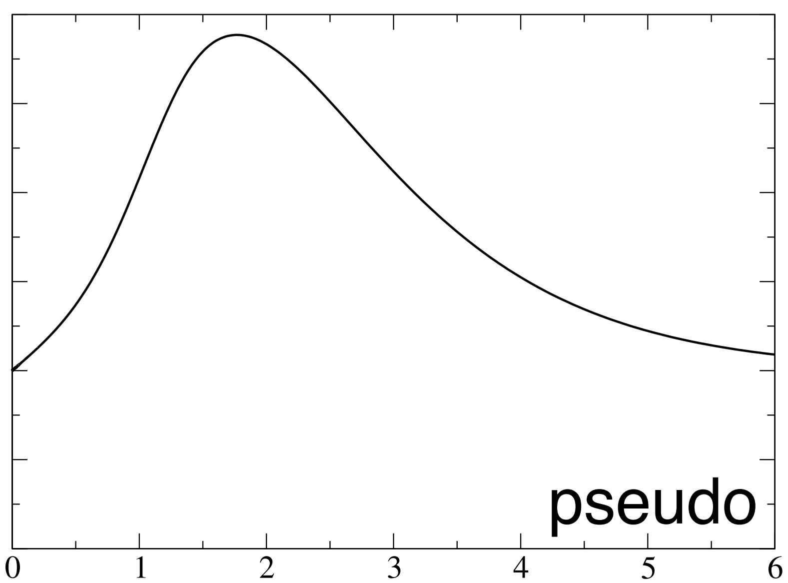
How to verify the precision of density-functional-theory implementations via reproducible and universal workflows. E. Bosoni et al. Nature Reviews Physics volume 6, pages 45–58 (2024)

Projector Augmented Waves

$$|\Psi\rangle = \mathcal{T}|\tilde{\Psi}\rangle$$

all-electron pseudo

$$\mathcal{T} = \mathbf{1} + \sum_{\mathbf{R},n} [|\phi_{\mathbf{R},n}\rangle - |\tilde{\phi}_{\mathbf{R},n}\rangle]\langle \tilde{p}_{\mathbf{R},n}|$$



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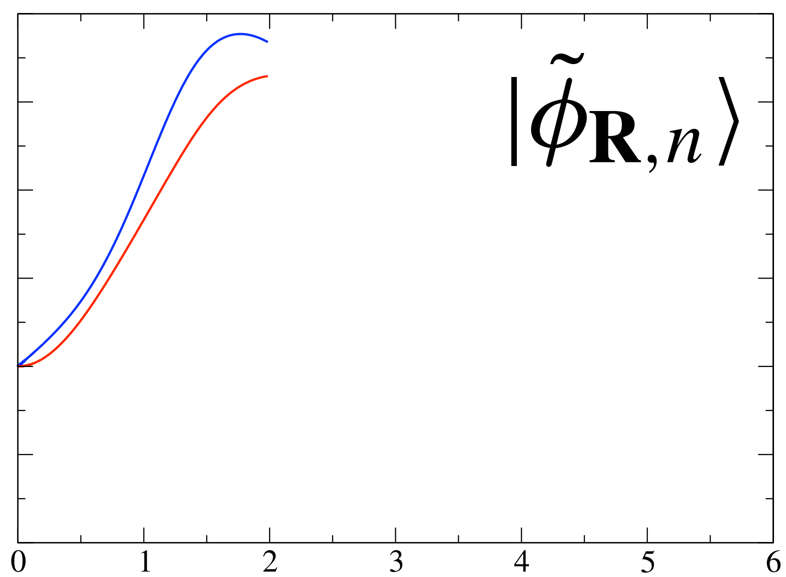

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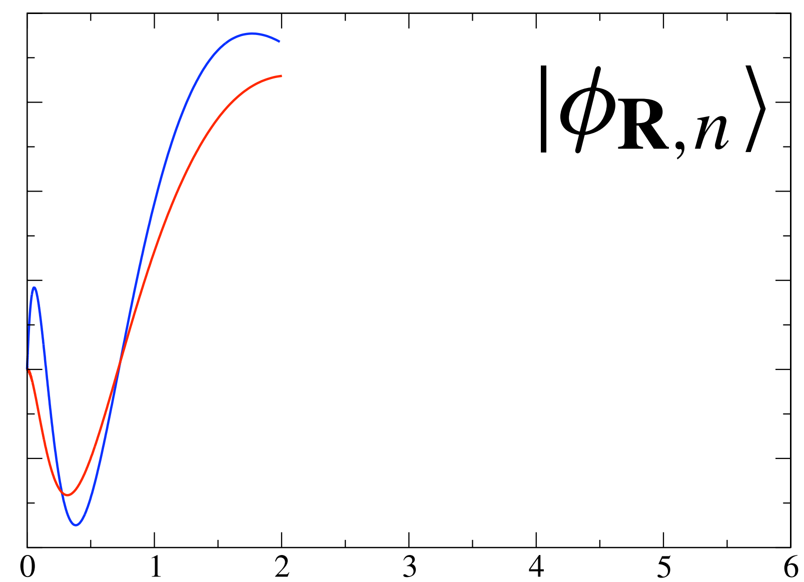

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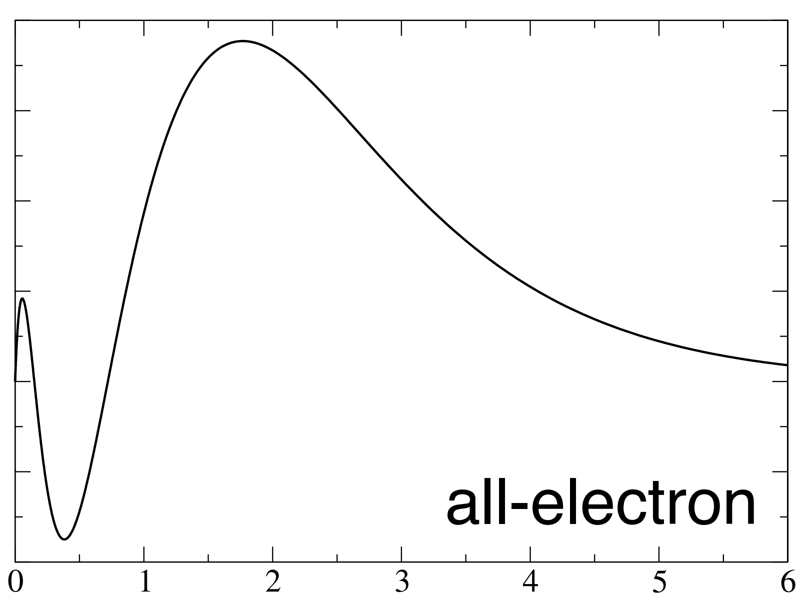
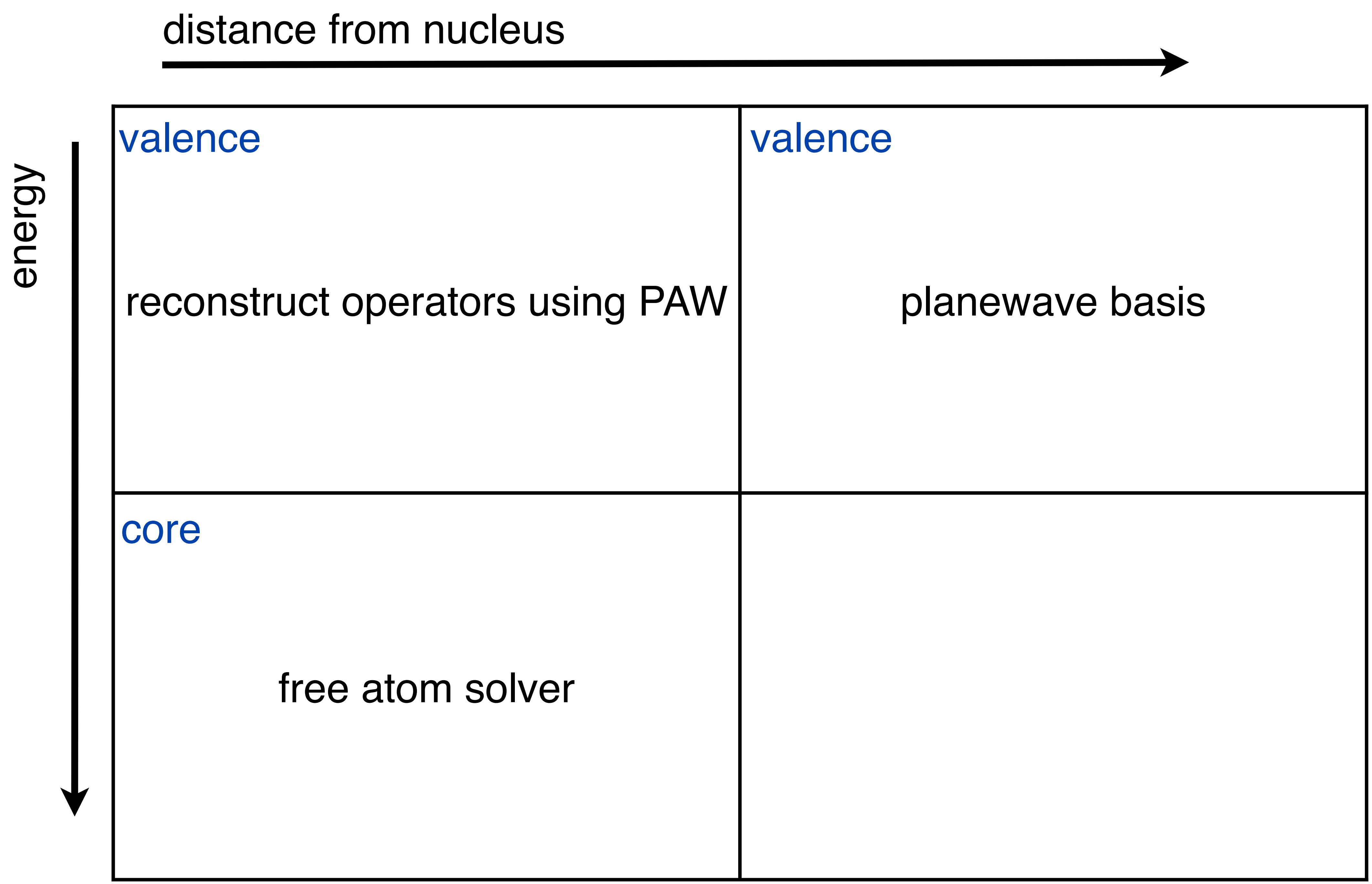


GIPAW

Gauge-Including Projector Augmented Waves
Modification of PAW by Pickard and Mauri for systems in an external magnetic field - plays a role similar to GIAO in quantum chemistry techniques

PAW / Pseudopotential Schematic



GIPAW

A theory for solid-state NMR

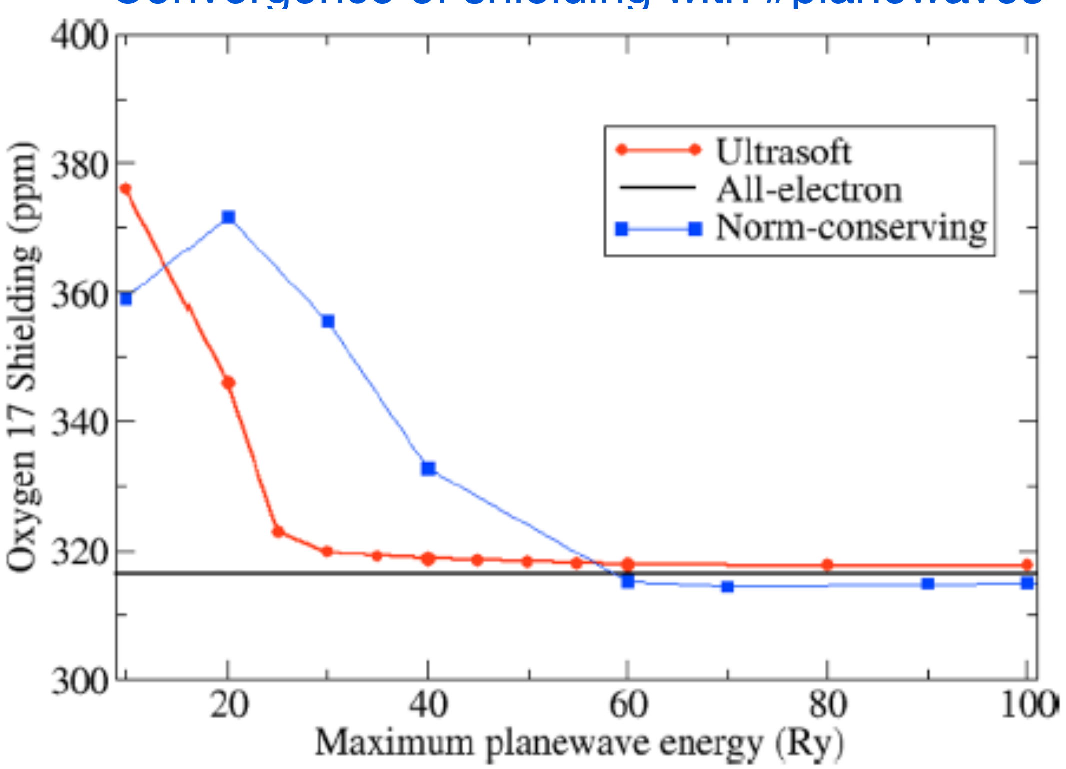
GIPAW vs Gaussian

test on small molecules

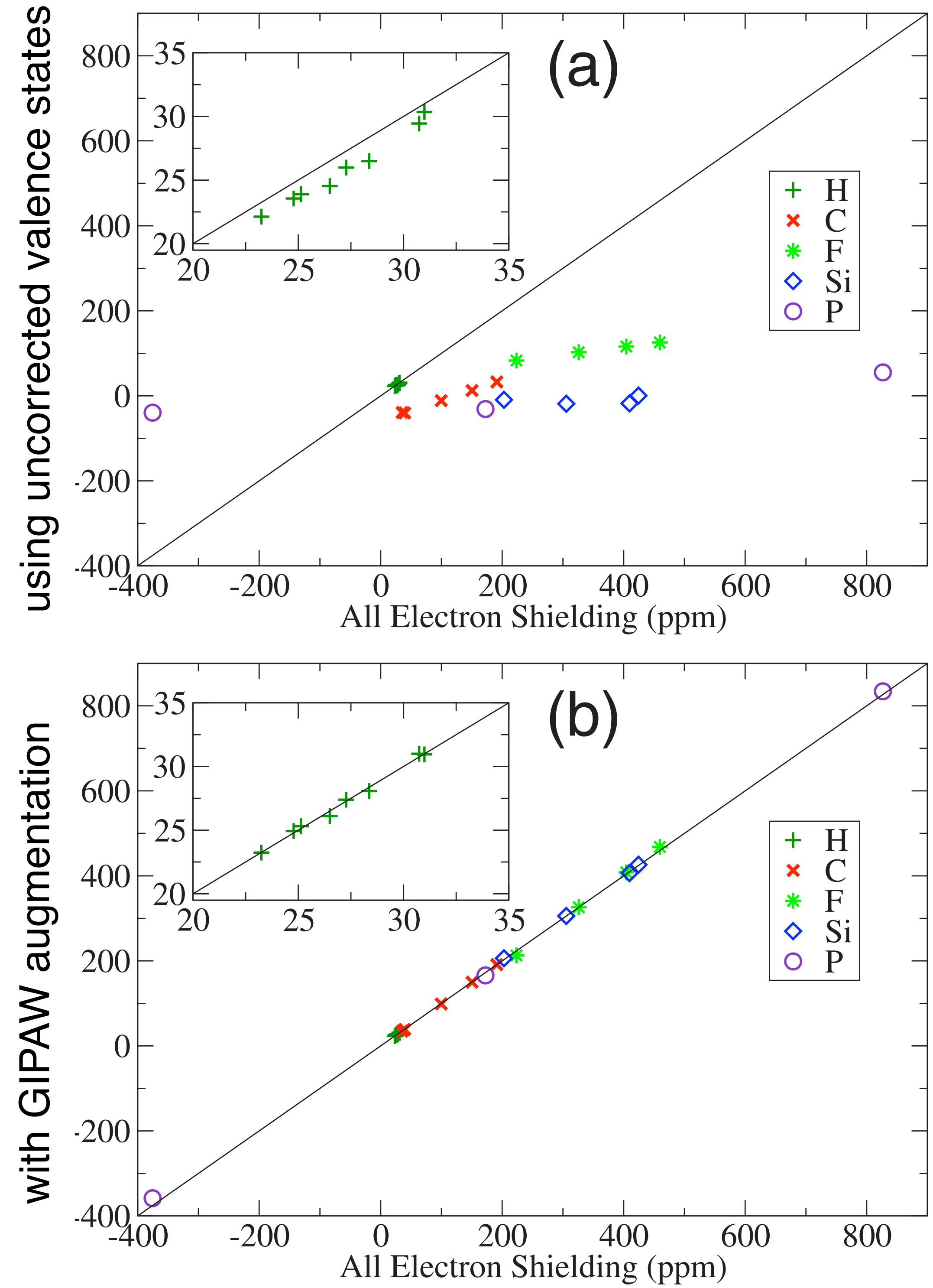
n.b. v. big Gaussian basis sets

JRY, C. Pickard, F. Mauri PRB 76, 024401 (2007)

Convergence of shielding with #planewaves



Chem. Rev. 112 (11), 5733-5779 (2012)

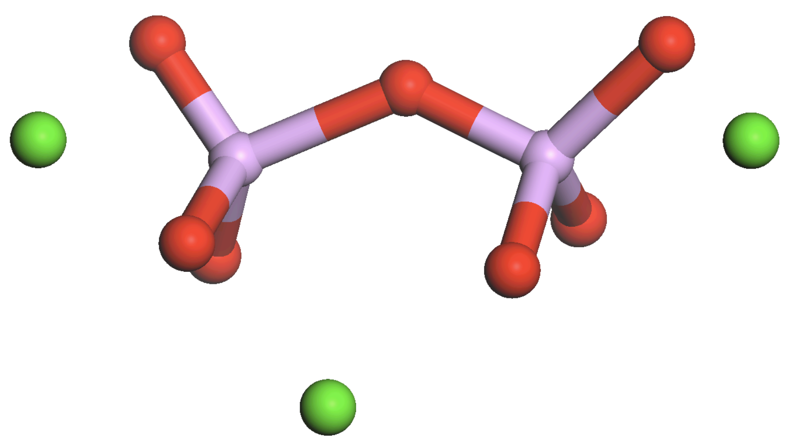


Materials Modelling Laboratory

OxfordMaterials

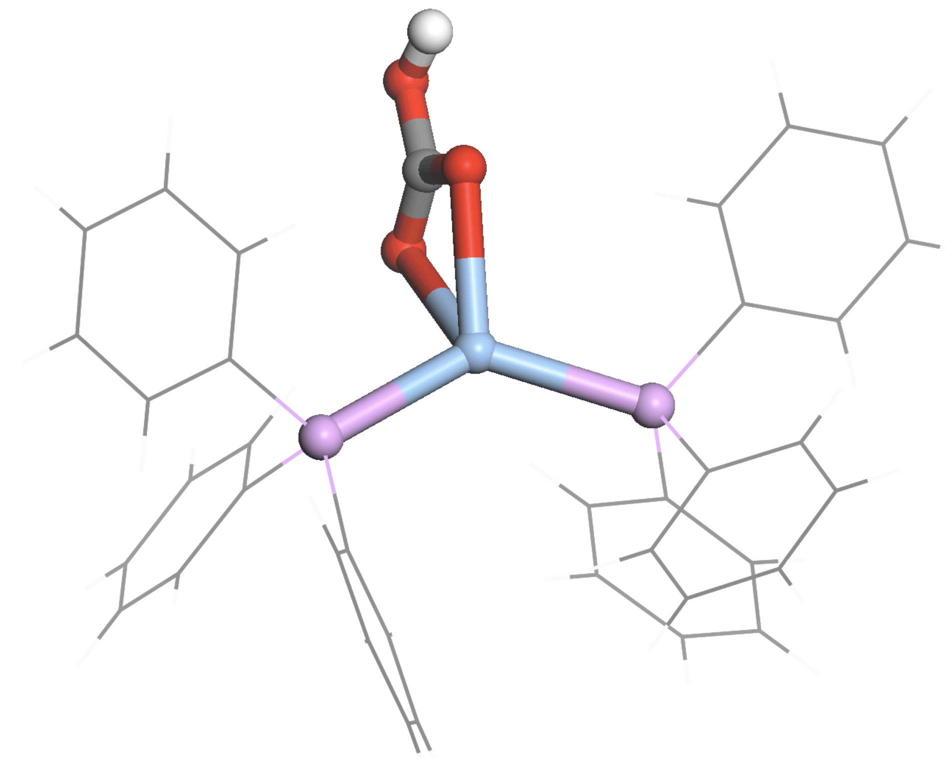
Phosphorus Couplings

α-$Ca_2P_2O_7$



	Expt	Calc
$^2J_{POP}$	14Hz	12Hz
$^1J_{PO}$	96Hz	105Hz

$AgP_2O_3CH(C_6H_5)_6$



	Expt	Calc
$^2J_{PAgP}$	147Hz	107Hz
$^1J_{P\text{-}Ag}$	469Hz	344Hz

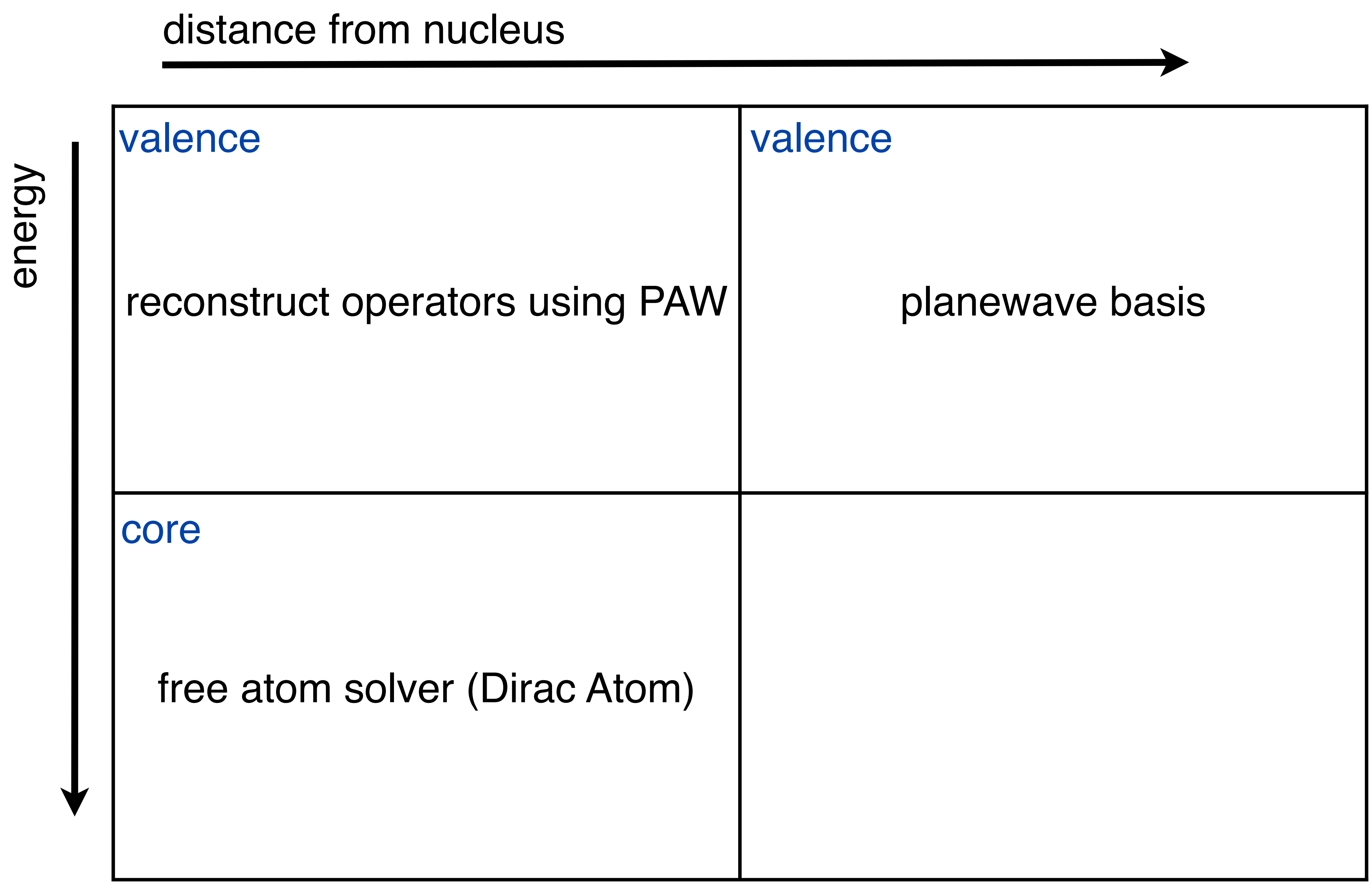
Dirac Equation

Core electrons are tightly bound => high kinetic energy

$$[c\alpha \cdot \hat{\mathbf{p}} + \beta c^2 + V]\psi = E\psi$$

Construct pseudopotential from Dirac Atomic calculation
scalar relativistic (average over j=1/2 & j=3/2 etc)
causes shift in energy of valence states

PAW / Pseudopotential Schematic



Dirac Equation

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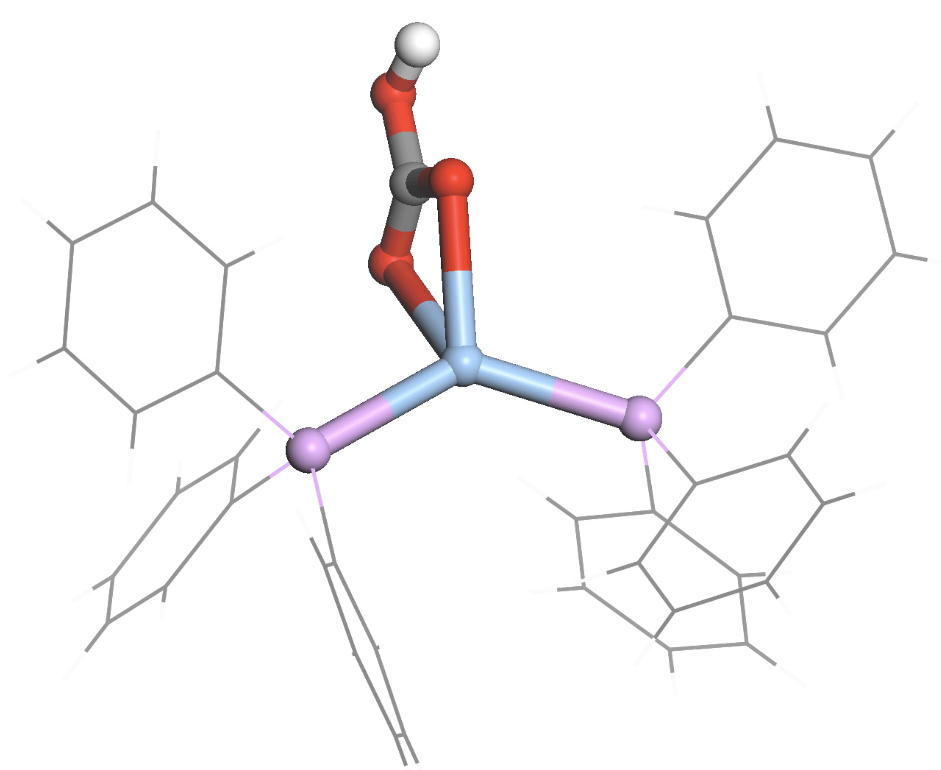
Construct pseudopotential from Dirac Atomic calculation
scalar relativistic (average over j=1/2 & j=3/2 etc)



$AgP_2O_3CH(C_6H_5)_6$

	Expt	non-rel	rel
$^2J_{PAgP}$	147Hz	107Hz	145Hz
$^1J_{P\text{-}Ag}$	475Hz	344Hz	------

Heavy atoms - relativistic effects

Indirect Effect

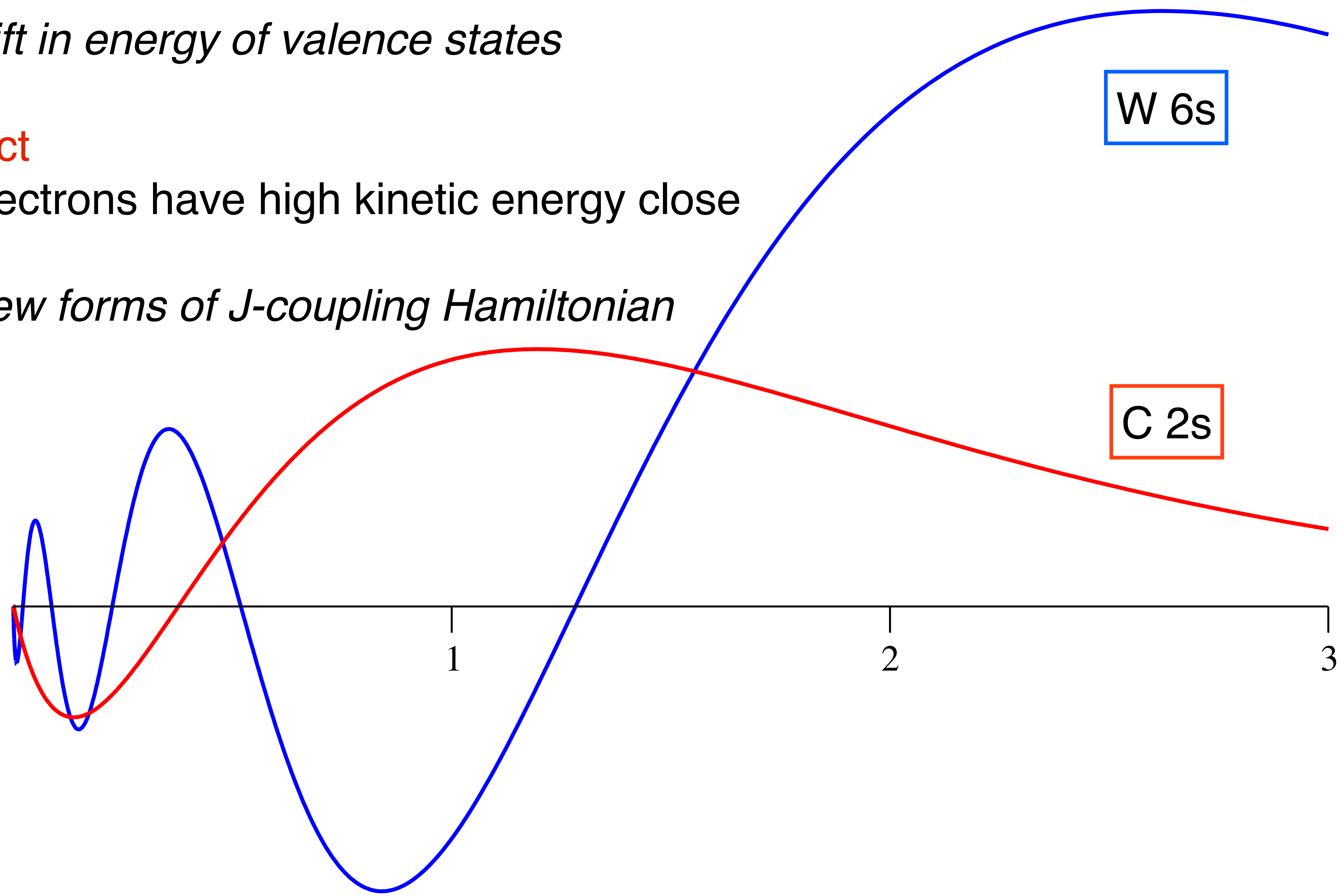
Core electrons are tightly bound => high kinetic energy

causes shift in energy of valence states

Direct Effect

Valence electrons have high kinetic energy close to nucleus

requires new forms of J-coupling Hamiltonian



Approximations to the Dirac Equation

$$\begin{pmatrix} V(\mathbf{r}) & c\,\boldsymbol{\sigma}\cdot\boldsymbol{\pi} \\ c\,\boldsymbol{\sigma}\cdot\boldsymbol{\pi} & V(\mathbf{r})-2c^2 \end{pmatrix}\begin{pmatrix} |\Phi\rangle \\ |\chi\rangle \end{pmatrix} = E\begin{pmatrix} |\Phi\rangle \\ |\chi\rangle \end{pmatrix}$$

$\downarrow$

Reduce to single spinor wavefunction

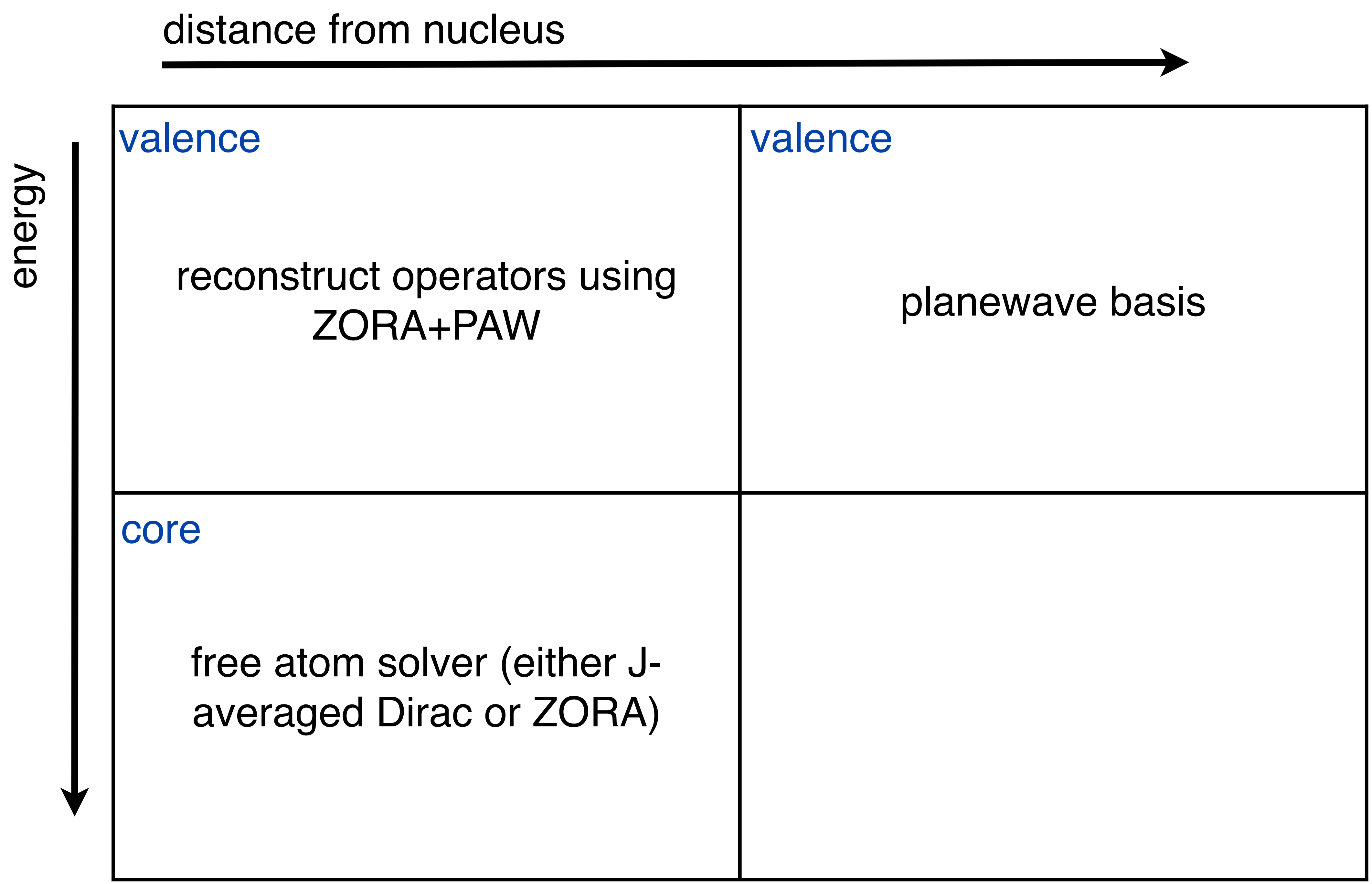
$p^2 \ll 4c^2$ $\rightarrow$ Pauli Equation

$E \ll (2c^2 - V(\mathbf{r}))$ $\rightarrow$ ZORA Equation

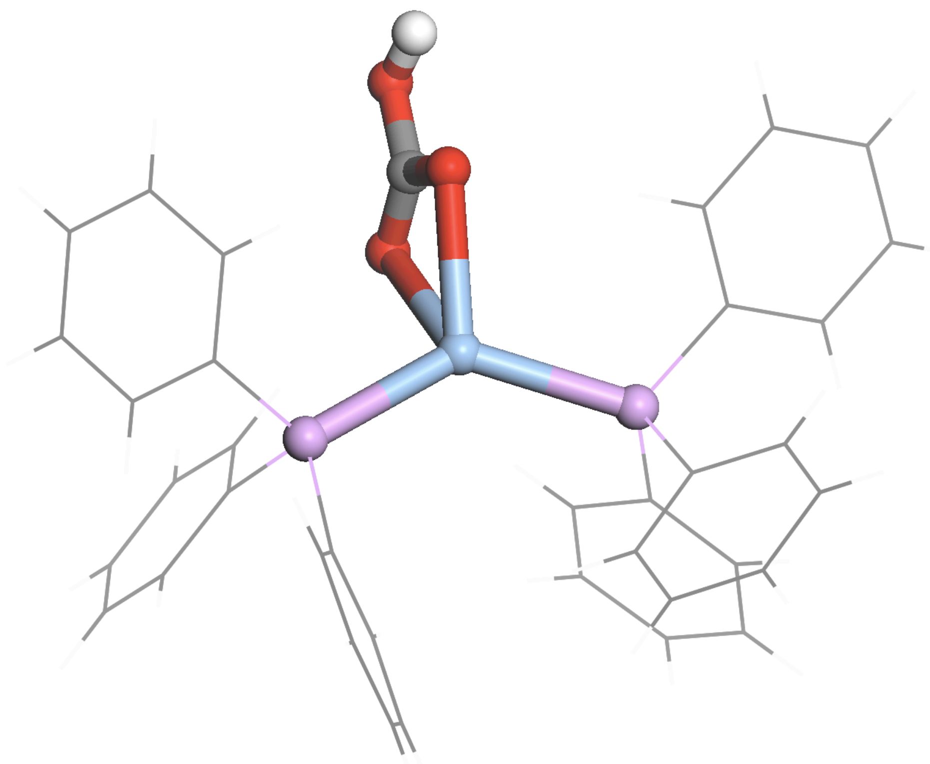
ZORA Equation

- Valid for a divergent potential (eg Coloumb)
- good approximation for low energy (valence states)
- less good for low lying core states

PAW / Pseudopotential Schematic



$AgP_2O_3CH(C_6H_5)_6$



	Expt	non-rel	rel
$^2J_{PAgP}$	147Hz	107Hz	145Hz
$^1J_{P\text{-}Ag}$	475Hz	344Hz	464Hz
$\Delta^1J_{P\text{-}Ag}$		120Hz	150Hz

$^2J_{PAgP}$ indirect effect

$^1J_{P\text{-}Ag}$ direct & indirect effect

Spin-orbit coupling

Condensed Matter Physics

Semiconductor band structures
Systems with non-collinear magnetism

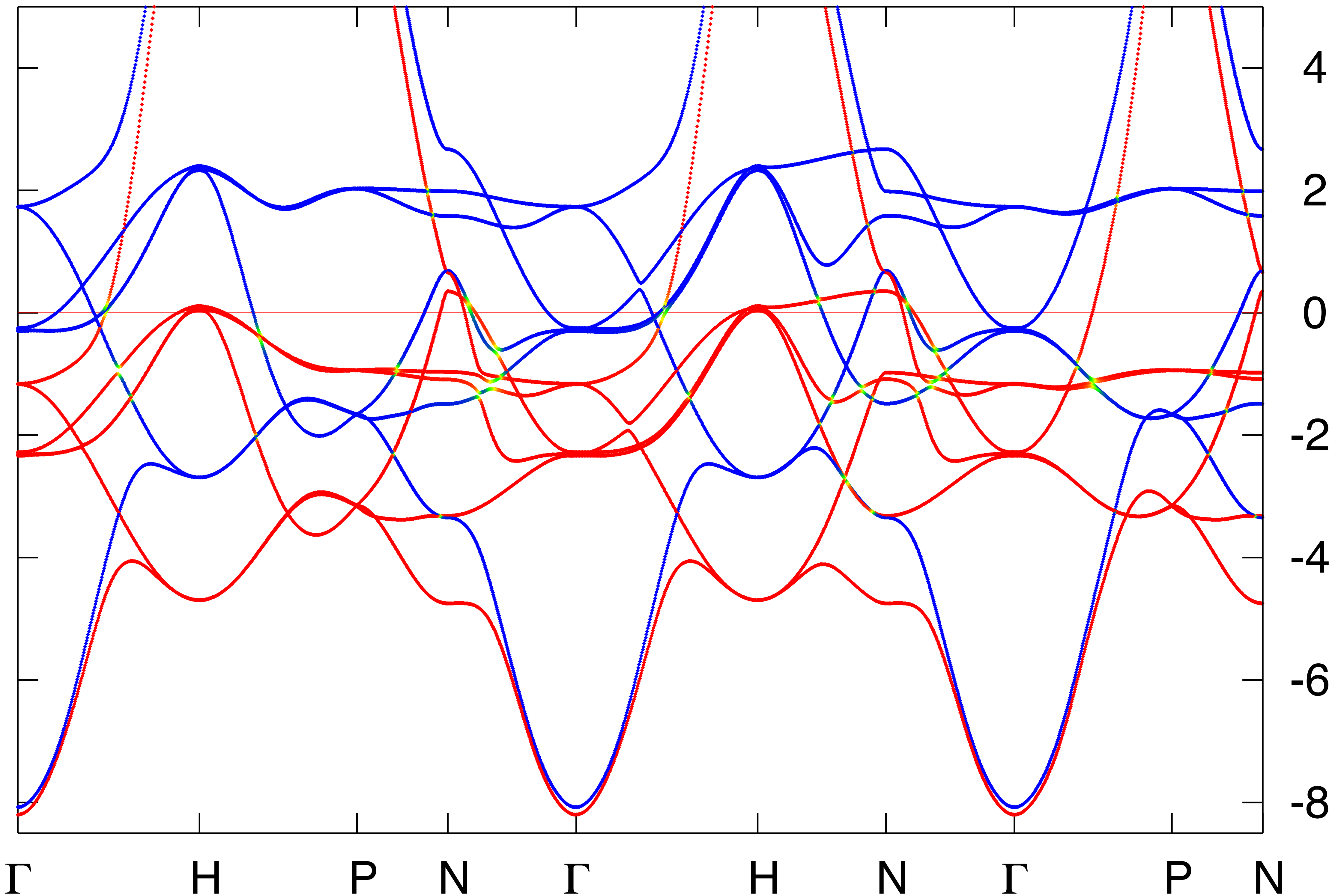
NMR

Chemical Shielding of light atoms bonded to heavy atoms
J-coupling between two heavy elements

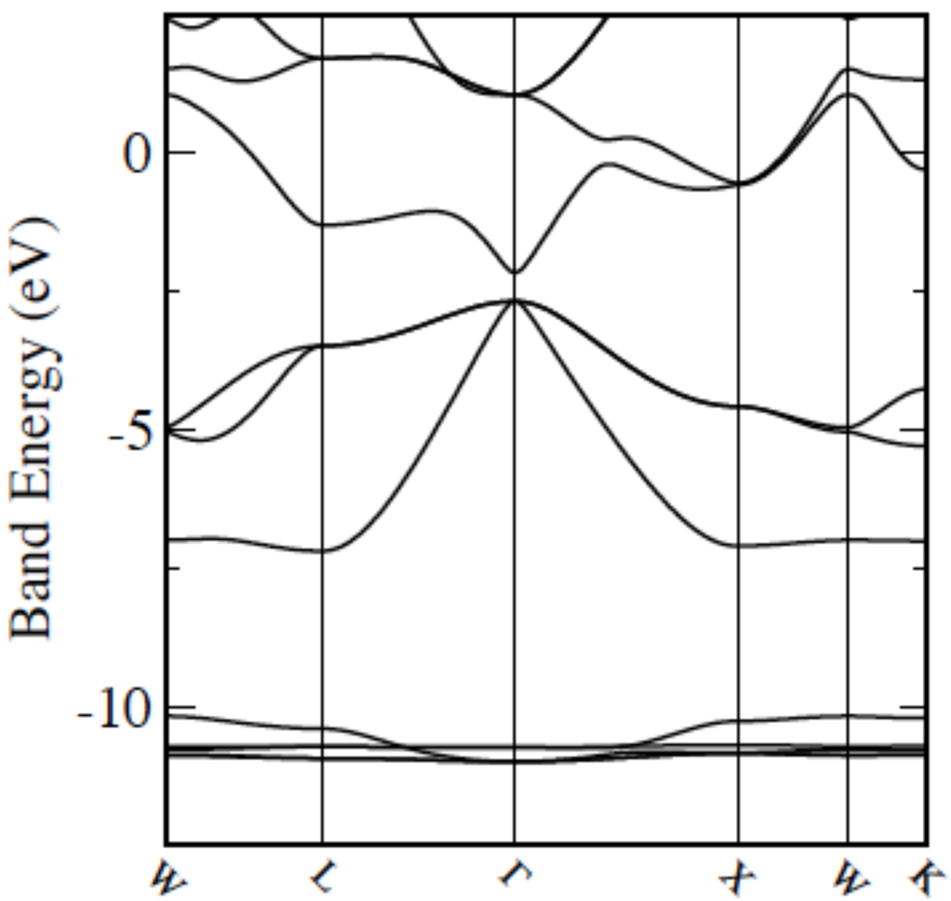
Including spin-orbit coupling in the planwave-pseudopotential approach

Construct Pseudopotential from Dirac Atom
Valence electrons experience a j-dependent potential
Valence wavefunctions described in spinor basis

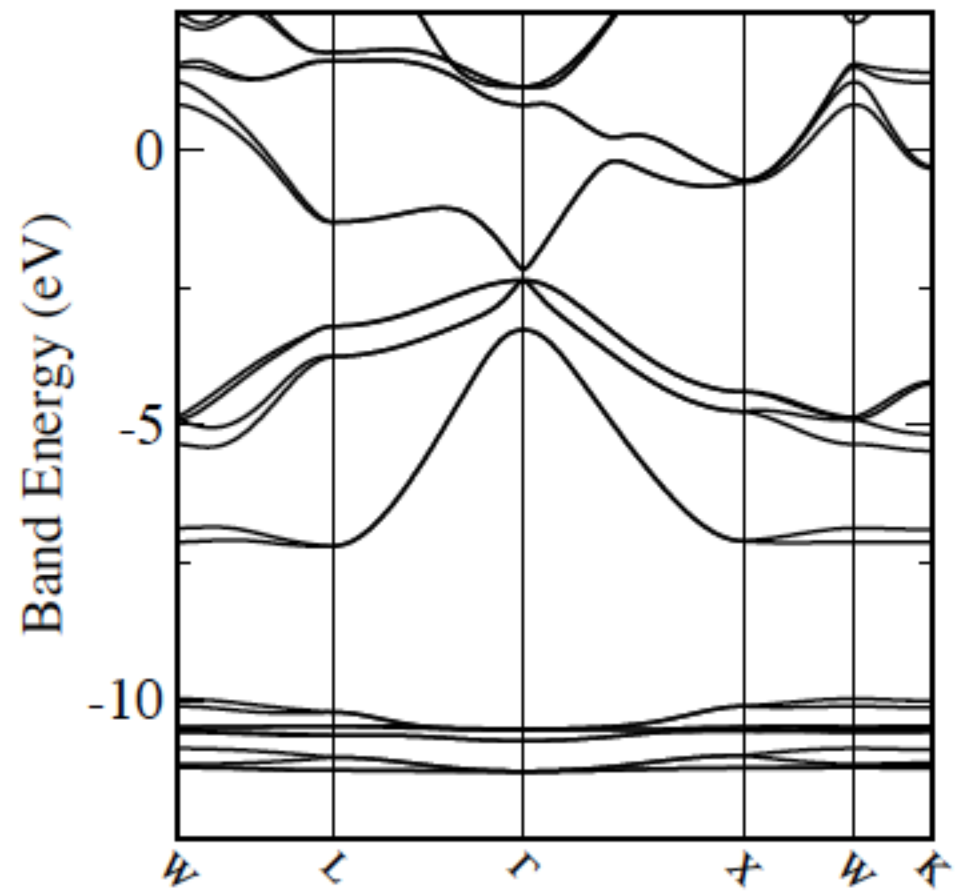
bcc - iron



CdTe



scalar



spin-orbit

Total energy, forces, stresses implemented in CASTEP
cost ~4x scalar

Limitations and Keywords

Linear response Phonons, Berry Phase, exact exchange functionals require norm-conserving potentials

NMR and Core-loss properties require on the fly potentials (as enables castep to automatically use (GI)PAW)

Spin-orbit needs J-dependant potentials (NCP only - use SOC set)

To control relativistic effects (on the fly generation only)

```
relativistic_treatment    = Koelling-Harmon (default - keep with this for any structure properties)

                          = Schrodinger (no relativistic effect - e..g to compare to Gaussian

                          = ZORA (scalar relativistic - most useful for NMR properties)
```

To turn on spin-orbit coupling

```
spin_treatment      =  VECTOR
spin_orbit_coupling =  true
```

```
Only energies, forces, stresses implemented
```

```
=============================================================
| Pseudopotential Report - Date of generation 31-07-2017    |
-------------------------------------------------------------
| Element: U Ionic charge: 14.00 Level of theory: PBESOL    |
| Atomic Solver: Dirac (FR)                                 |
|                                                           |
|               Reference Electronic Structure              |
|         Orbital         Occupation         Energy         |
|          6s1/2            2.000            -1.751          |
|          6p1/2            2.000            -1.101          |
|          6p3/2            4.000            -0.774          |
|          7s1/2            2.000            -0.157          |
|          5f5/2            1.286            -0.137          |
|          5f7/2            1.714            -0.106          |
|          6d3/2            0.400            -0.098          |
|          6d5/2            0.600            -0.080          |
|                                                           |
|                Pseudopotential Definition                 |
|        Beta      l      e       Rc      scheme   norm     |
|          1       0   -1.751   2.106     qc       1        |
|          2       0   -0.157   2.106     qc       1        |
|          3       1   -0.774   2.106     qc       1        |
|          4       1   -1.101   2.106     qc       1        |
|          5       2   -0.080   2.106     qc       1        |
|          6       2   -0.098   2.106     qc       1        |
|          7       3   -0.106   2.106     qc       1        |
|          8       3   -0.137   2.106     qc       1        |
|         loc      4    0.000   2.106     pn       0        |
|                                                           |
| No charge augmentation                                    |
| Partial core correction Rc = 1.472                        |
-------------------------------------------------------------
| "4|2.1|17|19|22|60N:70N:61N:62N:53N(qc=6,q3=8)"           |
-------------------------------------------------------------
```